MA 563 HW 1 Caiwei Zhang (cz239) Jan 21, 2026

1. For a matrix $A \in \mathbb{R}^{m\times n}$, $m \geq n$, let $\sigma_1, \sigma_2, \ldots, \sigma_r$ be the singular values of A and $\Lambda = \max_j \sigma_j$, $\lambda = \min_j \sigma_j$.

   1. Define the singular value decomposition of A.

   2. Prove that $\|A\|_2 \leq \|A\|_F$.

   3. Prove that $\lambda\|\mathrm{x}\|_2 \leq \|A\mathrm{x}\|_2 \leq \Lambda\|\mathrm{x}\|_2$ for all $\mathrm{x} \in \mathbb{R}^n$.

1. Ref: book property 1.7.

For $A \in \mathbb{R}^{m\times n}$, $m \geq n$, the SVD is a factorization of the form: $A = U\Sigma V^T$

where U is an $m\times m$ orthogonal matrix. The columns of U are called the left singular vectors of A.

V is an $n\times n$ orthogonal matrix. the columns of V are called the right singular vectors of A.

Σ is an $m\times n$ rectangular diagonal matrix. The diagonal entries $\Sigma_{ii} = \sigma_i$ are non-negative real numbers, i.e. the singular values of A.
(singular values are listed in descending order: $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$.).

2. $\|A\|_2 = \max\limits_{x\neq 0} \dfrac{\|Ax\|_2}{\|x\|_2} = \sigma_1$

$\|A\|_F = \sqrt{\sum_{i=1}^{n} \sigma_i^2} = \sqrt{\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_n^2}$

$\Rightarrow \|A\|_F^2 = \sigma_1^2 + \sigma_2^2 + \cdots + \sigma_n^2$

Since $\sigma_i^2 \geq 0 \ \forall i$, $\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_n^2 \geq \sigma_1^2$

$\|A\|_F^2 \geq \sigma_1^2$

$\|A\|_F \geq \sigma_1 \Rightarrow \|A\|_F \geq \|A\|_2$ $\square$.

3. Let $A = U\Sigma V^T$.

Then $\|Ax\|_2 = \|U\Sigma V^T x\|_2$.

$= \|U(\Sigma V^T x)\|_2 = \|\Sigma V^T x\|_2$ since U is an orthogonal matrix.

Let $y = V^T x$. Since V^T is orthogonal. $\|y\|_2 = \|x\|_2$.

$\|\Sigma y\|_2^2 = \sum_{i=1}^{n} (\sigma_i y_i)^2 = \sigma_1^2 y_1^2 + \sigma_2^2 y_2^2 + \cdots + \sigma_n^2 y_n^2$

Since $\Lambda = \max_i(\sigma_i)$, $\sigma_i \leq \Lambda \ \forall i$.

$\|Ax\|_2^2 = \sum_{i=1}^{n} \sigma_i^2 y_i^2 \leq \sum_{i=1}^{n} \Lambda^2 y_i^2 = \Lambda^2 \sum_{i=1}^{n} y_i^2 = \Lambda^2 \|y\|_2^2$

$\Rightarrow \|Ax\|_2 \leq \Lambda\|y\|_2$

Since $\lambda = \min_i(\sigma_i)$, $\sigma_i \geq \lambda \ \forall i$

$\sum_{i=1}^{n} \sigma_i^2 y_i^2 \geq \sum_{i=1}^{n} \lambda^2 y_i^2 = \lambda^2 \sum_{i=1}^{n} y_i^2 = \lambda^2 \|y\|_2^2$

$\Rightarrow \|Ax\|_2 \geq \lambda\|y\|_2$

Since $\|y\|_2 = \|x\|_2$, $\lambda\|x\|_2 \leq \|Ax\|_2 \leq \Lambda\|x\|_2$. $\square$.

chp 1

12. Let $A \in \mathbb{C}^{m\times n}$ with $\text{rank}(A) = n$. Show that $A^\dagger = (A^T A)^{-1} A^T$ enjoys the following properties:

$$(1)\ A^\dagger A = I_n;\ (2)\ A^\dagger A A^\dagger = A^\dagger,\ A A^\dagger A = A;\ (3)\ \text{if } m = n,\ A^\dagger = A^{-1}.$$

(1) $A^\dagger A = [(A^TA)^{-1}A^T]A$

$= (A^TA)^{-1}(A^TA)$ by association.

For any invertible matrix X, $X^{-1}X = I$.

so. $(A^TA)^{-1}(A^TA) = I$.

i.e. $A^\dagger A = I$.

(2) $A^\dagger A A^\dagger = I A^\dagger = A^\dagger$ $\square$.

by (1).

(3) if $m=n$. A is a square matrix.

since $\text{rank}(A)=n$. A is nonsingular. (invertible)

so A^T is also invertible.

so $(A^TA)^{-1} = A^{-1}(A^T)^{-1}$

so $A^\dagger = (A^TA)^{-1}A^T$

$= A^{-1}\underbrace{(A^T)^{-1}A^T}_{= I} = A^{-1}$ $\square$.

chp 2.

2. Study the well posedness and the conditioning in the infinity norm of the following problem as a function of the datum d: find x and y such that

$$\begin{cases} x + dy = 1, \\ dx + y = 0. \end{cases}$$

In form $Ax=b$:

$$\underbrace{\begin{bmatrix} 1 & d \\ d & 1 \end{bmatrix}}_{A} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$\det(A) = 1-d^2$.

if $\det(A) \neq 0$. A is invertible and. the problem is well posed.

$1-d^2 \neq 0$

$d \neq \pm 1$

so the problem is well posed $\forall\, d$ s.t. $d \neq 1$ and $d \neq -1$.

For conditioning, $K_\infty(A) = \|A\|_\infty \|A^{-1}\|_\infty$.

$\Rightarrow \|A\|_\infty = 1+|d|$. $\quad A^{-1} = \frac{1}{\det(A)}\begin{bmatrix} 1 & -d \\ -d & 1 \end{bmatrix} = \frac{1}{1-d^2}\begin{bmatrix} 1 & -d \\ -d & 1 \end{bmatrix}$

$\|A^{-1}\|_\infty = \left|\frac{1}{1-d^2}\right| + \left|\frac{-d}{1-d^2}\right| = \frac{1+|d|}{|1-d^2|}$

$\Rightarrow K_\infty(A) = (1+|d|)\cdot\frac{1+|d|}{|1-d^2|} = \frac{(1+|d|)^2}{|1-d^2|} = \left|\frac{|d|+1}{|d|-1}\right|$

Therefore, the problem is ill-conditioned as $|d| \to 1$ because $\kappa_\infty(A) \to \infty$.
This aligns with the fact that problem is not well-posed at $d = \pm 1$

3. Study the conditioning of the solving formula $x_\pm = -p \pm \sqrt{p^2+q}$ for the second degree equation $x^2 + 2px - q$ with respect to changes in the parameters p and q separately.

By 2.19. Relative condition, $K(d)$ for a function $G(d)$ measures the sensitivity of the output to small changes in the input d.

$$K(d) \approx \left| \frac{d \cdot G'(d)}{G(d)} \right|$$

Given $x = -p \pm (p^2+q)^{1/2}$.

$\boxed{\text{For } p}$

$$\frac{\partial x}{\partial p} = -1 \pm \frac{1}{2}(p^2+q)^{-1/2} \cdot (2p)$$

$$\frac{\partial x}{\partial p} = -1 \pm \frac{p}{\sqrt{p^2+q}} = \frac{-\sqrt{p^2+q} \pm p}{\sqrt{p^2+q}}$$

For $x_+ = -p + \sqrt{p^2+q}$.

Numerator $= -\sqrt{p^2+q} + p = -(-p + \sqrt{p^2+q}) = -x_+$

$$\frac{\partial x_+}{\partial p} = \frac{-x_+}{\sqrt{p^2+q}}$$

For $x_- = -p - \sqrt{p^2+q}$

Numerator $= -\sqrt{p^2+q} + q = x_-$

$$\frac{\partial x_-}{\partial p} = \frac{x_-}{\sqrt{p^2+q}}$$

so $K(p) = \left| \frac{p}{x} \cdot \frac{\partial x}{\partial p} \right|$

$$= \left| \frac{p}{x} \cdot \frac{\pm x}{\sqrt{p^2+q}} \right| = \left| \frac{\pm p}{\sqrt{p^2+q}} \right| = \frac{|p|}{\sqrt{p^2+q}}.$$

$\boxed{\text{For } q}$

$$\frac{\partial x}{\partial q} = 0 \pm \frac{1}{2}(p^2+q)^{-1/2} \cdot (1)$$

$$= \pm \frac{1}{2\sqrt{p^2+q}}$$

so $K(q) = \left| \frac{q}{x} \cdot \frac{\partial x}{\partial q} \right|$

$$= \left| \frac{q}{x} \cdot \frac{\pm 1}{2\sqrt{p^2+q}} \right| = \frac{|q|}{2|x|\sqrt{p^2+q}}$$

10. Prove (2.37).

WTS: $\dfrac{|x \oplus y - (x+y)|}{|x+y|} \le u(1+u)\dfrac{|x|+|y|}{|x+y|} + u.$ (2.37).

by triangle inequality, $|x \oplus y - (x+y)| = |x \oplus y - (fl(x)+fl(y)) + (fl(x)+fl(y)) - (x+y)|$

$$\le \underbrace{|x \oplus y - (fl(x)+fl(y))|}_{\text{operation error}} + \underbrace{|(fl(x)-x)+(fl(y)-y)|}_{\text{representation error}}$$

From property 2.1. the floating point representation of a real number x is given by $fl(x) = x(1+\delta)$ with $|\delta| \le u$.

Thus, $fl(x) - x = x\delta_x$ with $|\delta_x| \le u$.

$fl(y) - y = y\delta_y$ with $|\delta_y| \le u$.

So $|(fl(x)-x)+(fl(y)-y)| = |x\delta_x + y\delta_y| \le |x||\delta_x| + |y||\delta_y| \le (|x|+|y|)u.$

Then, by 2.36, for any $a, b \in \mathbb{F}$, $fl(a+b) = (a+b)(1+\delta_1)$ with $|\delta_1| \le u$.

Let $a = fl(x)$ and $b = fl(y)$.

Then $x \oplus y = (fl(x)+fl(y))(1+\delta_1)$

$$\Rightarrow |x \oplus y - (fl(x)+fl(y))| = |(fl(x)+fl(y))(1+\delta_1) - (fl(x)+fl(y))|$$
$$= |(fl(x)+fl(y))\delta_1|$$
$$= |(x + x\delta_x + y + y\delta_y)\delta_1|$$

By triangle inequality and the bounds $|\delta| \le u$.

$$\le (|x+y| + |x|u + |y|u)u.$$
$$= |x+y|u + (|x|+|y|)u^2.$$

Therefore, $|x \oplus y - (x+y)| \le [|x+y|u + (|x|+|y|)u^2] + [(|x|+|y|)u].$

$$= |x+y|u + u(1+u)(|x|+|y|)$$

so relative error

$$\frac{|x \oplus y - (x+y)|}{|x+y|} \le \frac{|x+y|u}{|x+y|} + \frac{u(1+u)(|x|+|y|)}{|x+y|}$$

$$\frac{|x \oplus y - (x+y)|}{|x+y|} \le u + u(1+u)\frac{|x|+|y|}{|x+y|} \qquad \square.$$

Chp 8.

3. Prove that $\omega'_{n+1}(x_i) = \prod_{\substack{j=0 \\ j\neq i}}^{n}(x_i - x_j)$ where ω_{n+1} is the nodal polynomial (8.6).

Then, check (8.5).

By 8.6, The nodal polynomial is defined as

$$w_{n+1}(x) = \prod_{k=0}^{n}(x-x_k) = (x-x_0)(x-x_1)\cdots(x-x_n)$$

Let polynomial $g_i(x) = \prod_{\substack{j=0 \\ j\neq i}}^{n}(x-x_j)$ (product of all except the i-th one)

so $w_{n+1}(x) = (x-x_i)\,g_i(x)$

$$w'_{n+1}(x) = \frac{d}{dx}\left[(x-x_i)\right]g_i(x) + (x-x_i)\,g'_i(x)$$

since $\frac{d}{dx}(x-x_i) = 1$

$$w'_{n+1}(x) = 1\cdot g_i(x) + (x-x_i)\,g'_i(x)$$

$$w'_{n+1}(x_i) = 1\cdot g_i(x_i) + 0 = g_i(x_i).$$

Therefore, $w'_{n+1}(x_i) = \prod_{\substack{j=0 \\ j\neq i}}^{n}(x-x_j)$ $\square$.

Check 8.5: $\pi_n(x) = \sum_{i=0}^{n}\frac{w_{n+1}(x)}{(x-x_i)w'_{n+1}(x_i)}\,y_i$

From 8.3 and 8.4. the standard Lagrange form:

$$\pi_n(x) = \sum_{i=0}^{n} y_i\, l_i(x). \quad \text{where} \quad l_i(x) = \prod_{\substack{j=0 \\ j\neq i}}^{n}\frac{x-x_j}{x_i-x_j}$$

$w_{n+1}(x) \leftrightarrow l_i(x)$

$$\prod_{\substack{j=0 \\ j\neq i}}^{n}(x-x_j) = \frac{\prod_{j=0}^{n}(x-x_j)}{(x-x_i)} = \frac{w_{n+1}(x)}{x-x_i}$$

$w'_{n+1}(x) \leftrightarrow l_i(x)$

$$\prod_{\substack{j=0 \\ j\neq i}}^{n}(x_i-x_j) = w'_{n+1}(x_i)$$

$$l_i(x) = \frac{w_{n+1}(x)}{(x-x_i)\,w'_{n+1}(x_i)}$$

Substitute $l_i(x)$ to $\pi_n(x)$.

$$\pi_n(x) = \sum_{i=0}^{n} y_i\left[\frac{w_{n+1}(x)}{(x-x_i)\,w'_{n+1}(x_i)}\right] \quad (8.5)$$

$\square$.